



BY 512: Quantitative and Computational Biology

Course Code	BY 512
Course Name	Quantitative and Computational Biology
L-T-P-C	3-0-0-3
Prerequisites	IC136, or consent of faculty
Intended for	B. Tech. 3rd and 4th year, MS/M. Tech., Ph.D.
Distribution	Core for M. Tech. Biotechnology, elective for others
Approval	9th Senate (May 2016)

Course Contents

- **Quantitative Biology**

Probability Theory, Probability Distributions - Binomial, Gaussian and Poisson Distributions.

Descriptive statistics: mean, variance and sum of squares; mean and variance of a distribution, random numbers, random sampling.

Regression analysis: linear, multiple and nonlinear.

Test of hypotheses: t-test, z-test; Chi-square test of independence.

Multivariate Analysis: various types of classification, ANOVA, PCA Examples of Statistics in biological data analysis.[21 Lectures]

- **Computational biology: Bioinformatics, bio-algorithms and Tools**

Introduction to Basic Programming: Introduction to basic scripting and programming routinely used in computational biology.

Biological Databases and Sequence File Formats: Introduction to different biological databases, their classification schemes, and biological database retrieval systems.

Sequence Alignments: Introduction to concept of alignment, Scoring matrices, Alignment algorithms for pairs of sequences, Multiple sequence alignment.

Gene Prediction Methods: What is gene prediction? Computational methods of gene prediction-prokaryotic & eukaryotic.

Molecular Phylogeny: Introduction to phenotypic and molecular phylogeny. Representation of phylogeny, Molecular clocks, Methods of phylogenetic construction, statistical evaluation of the obtained phylogenetic trees.

Introduction to systems biology: Different Omics, Metabolic pathways and networks. [21 Lectures]

Text and Reference Books

1. Dekking, F.M., Kraaikamp, C., Lopushaä, H.P., Meester, L.E., **A Modern Introduction to Probability and Statistics - Understanding Why and How**, Springer.
2. Norman Bailey, **Statistical methods in Biology**, Latest edition.
3. S.C. Rastogi, N. Mendiratta, P. Rastogi, **Methods and Applications Genomics, Proteomics, and Drug Discovery**, 3rd Edition, PHI Learning Private Limited.
4. Z. Ghosh and B. Mallick, **Bioinformatics Principles and Applications**, Oxford University Press.
5. Arthur M. Lesk, **Introduction to Bioinformatics**, 3rd edition, Oxford University Press.
6. Latest research articles will be advised related to the topic being taught.